System Administration and Management

ET 8413

Unit I

System Administration & IT Infrastructure

Introduction

- **System administration refers to the management, configuration, maintenance, and troubleshooting of computer systems, servers, networks, and IT infrastructure to ensure their optimal performance, security, and availability.**
- **A system administrator (sysadmin) ensures that an organization's IT systems run smoothly, securely, and efficiently.**

Introduction

- System Comprise of
- Computers
- Networks
- Users
- Any thing else?

Duties of System Administrator

- Applying operating system updates, patches, and configuration changes.
- Installing and configuring new hardware and software.
- Adding, removing, or updating user account information, resetting passwords, etc.
- Responsibility for documenting the configuration of the system.
- Responsibility for security.

Duties of System Administrator

- Performing routine audits of systems and software.
- Performing backups.
- Analyzing system logs and identifying potential issues with computer systems.
- Troubleshooting any reported problems.
- Introducing and integrating new technologies into existing data center environments.
- Answering technical queries.

Types of Administrators

- In a larger company, following may all be separate positions within a computer support or Information Services (IS) department.
- In a smaller group they may be shared by a few sysadmins, or even a single person.
- A database administrator (DBA) maintains a database system, and is responsible for the integrity of the data and the efficiency and performance of the system.

Types of Administrators

- A network administrator maintains network infrastructure such as switches and routers, and diagnoses problems with these or with the behavior of network-attached computers.

- A security administrator is a specialist in computer and network security, including the administration of security devices such as firewalls, as well as consulting on general security measures.

Types of Administrators

- A **web administrator** maintains web server services (such Apache) that allow for internal or external access to web sites. Tasks include managing multiple sites, administering security, and configuring necessary components and software.
- **Technical support staff** respond to individual users' difficulties with computer systems, provide instructions and sometimes training, and diagnose problems.

Operating systems supporting Administration

- **System administrators work with various operating systems (OS) depending on the organization's needs. The most commonly used OS in system administration include:**

- ***Windows-Based Operating Systems:*** **Windows Server is widely used in corporate environments, especially where Active Directory (AD) and Microsoft-based services are required.**

Operating systems supporting Administration

- **Windows Server Versions for SysAdmins:**
- **Windows Server 2022 (Latest stable version)**
  - **Features: Active Directory, Hyper-V, PowerShell, Azure integration.**
- **Windows Server 2019 (Still widely used)**
- **Windows Server 2016 (Legacy support)**
- ***Linux Distributions (Open-Source & Enterprise):*** **Popular for web servers, cloud computing, and DevOps environments.**

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Operating systems supporting Administration

- **Popular Distros:**
  - **Ubuntu Server – User-friendly with strong community support**
  - **CentOS / Rocky Linux – Stable and widely used for hosting**
  - **Red Hat Enterprise Linux (RHEL) – Enterprise-grade with commercial support**
  - **Debian – Reliable and secure for long-term use**

Operating systems supporting Administration

- ***macOS in System Administration:*** **While not common for servers, macOS is used in:**
  - **Managing Apple devices in an enterprise (Jamf Pro, MDM)**
  - **DevOps & development environments (Unix-like terminal)**

System Administration Tools

- SysAdmins use a variety of tools to monitor, manage, and automate IT infrastructure.
- Wireshark – Packet capture and network analysis
- Nagios- Network & server monitoring (alerts for outages).
- Nessus- Vulnerability scanning.
- Nmap – Network scanning and security auditing

Physical Layer

- Provides physical interface for transmission of information.
- Defines rules by which bits are passed from one system to another on a physical communication medium.
- Covers all - mechanical, electrical, functional and procedural - aspects for physical communication.
- Such characteristics as voltage levels, timing of voltage changes, physical data rates, maximum transmission distances, physical connectors, and other similar attributes are defined by physical layer specifications.

Data Link Layer

- Data link layer attempts to provide reliable communication over the physical layer interface.

- Breaks the outgoing data into frames and reassemble the received frames.
- Create and detect frame boundaries.
- Handle errors by implementing an acknowledgement and retransmission scheme.
- Implement flow control.
- Supports points-to-point as well as broadcast communication.
- Supports simplex, half-duplex or full-duplex communication.

Network Layer

- Implements routing of frames (packets) through the network.
- Defines the most optimum path the packet should take from the source to the destination
- Defines logical addressing so that any endpoint can be identified.
- Handles congestion in the network.
- Facilitates interconnection between heterogeneous networks (Internetworking).
- The network layer also defines how to fragment a packet into smaller packets to accommodate different media.

Transport Layer

- Purpose of this layer is to provide a reliable mechanism for the exchange of data between two processes in different computers.

- Ensures that the data units are delivered error free.
- Ensures that data units are delivered in sequence.
- Ensures that there is no loss or duplication of data units.
- Provides connectionless or connection oriented service.
- Provides for the connection management.
- Multiplex multiple connection over a single channel.

Session Layer

- Session layer provides mechanism for controlling the dialogue between the two end systems. It defines how to start, control and end conversations (called sessions) between applications.

- This layer requests for a logical connection to be established on an end-user's request.
- Any necessary log-on or password validation is also handled by this layer.
- Session layer is also responsible for terminating the connection.
- This layer provides services like dialogue discipline which can be full duplex or half duplex.
- Session layer can also provide check-pointing mechanism such that if a failure of some sort occurs between checkpoints, all data can be retransmitted from the last checkpoint.

Presentation Layer

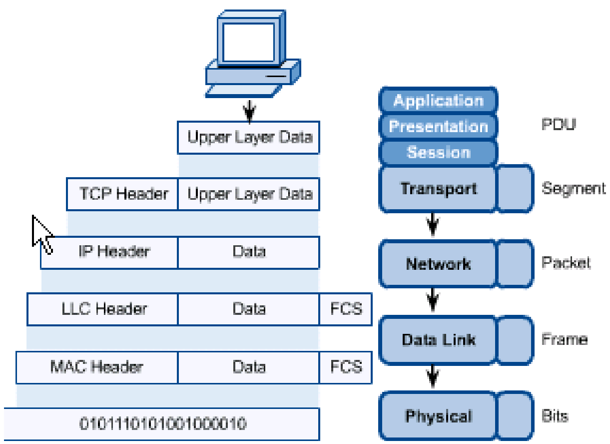
- Presentation layer defines the format in which the data is to be exchanged between the two communicating entities.
- Also handles data compression and data encryption (cryptography).

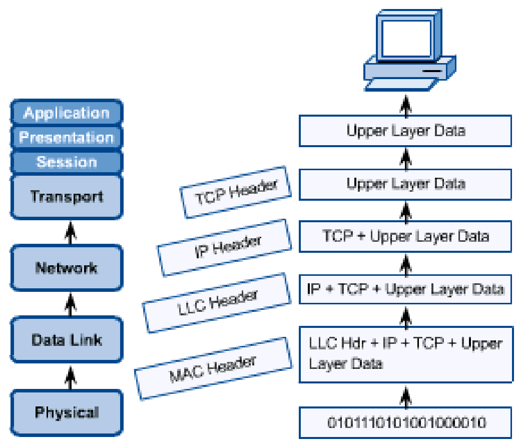
Application Layer

- Application layer interacts with application programs and is the highest level of OSI model.
- Application layer contains management functions to support distributed applications.
- Examples of application layer are applications such as file transfer, electronic mail, remote login etc.

OSI in Action

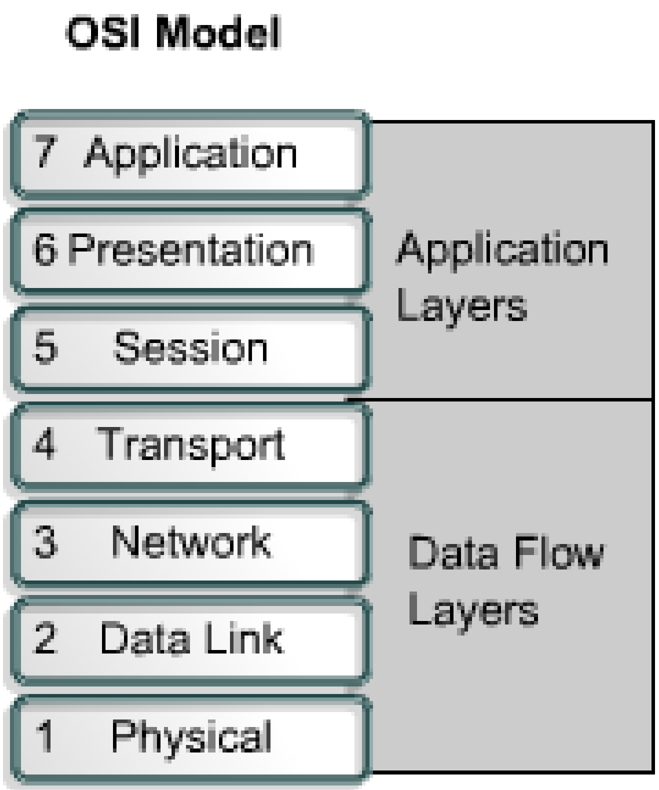
- A message begins at the top application layer and moves down the OSI layers to the bottom physical layer.
- As the message descends, each successive OSI model layer adds a header to it.
- A header is layer-specific information that basically explains what functions the layer carried out.
- Conversely, at the receiving end, headers are striped from the message as it travels up the corresponding layers.

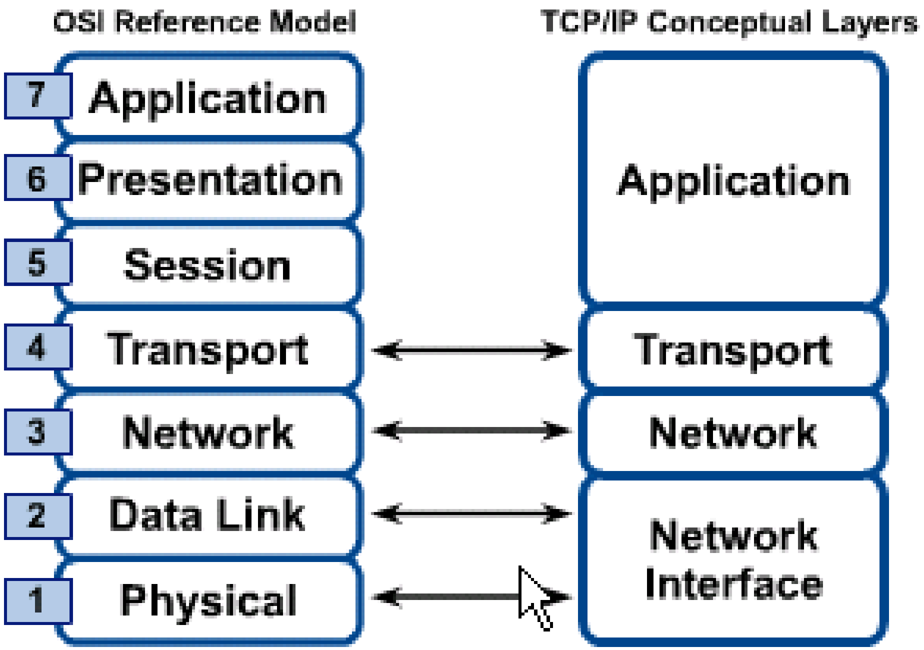




TCP/IP MODEL

OSI & TCP/IP Models





TCP/IP Model

Application Layer

Application programs using the network

Transport Layer (TCP/UDP)

Management of end-to-end message transmission, error detection and error correction

Network Layer (IP)

Handling of datagrams : routing and congestion

Data Link Layer

Management of cost effective and reliable data delivery, access to physical networks

Physical Layer

Physical Media